

JAMES LEINER

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EDUCATION

- Carnegie Mellon University** — Ph.D. in Statistics and Data Science 2021 – 2026 (expected)
Thesis: Advances in Human-in-the-Loop Inference
Advisor: Aaditya Ramdas
Committee: Larry Wasserman, Jing Lei, Lucas Janson, and Daniela Witten
- The University of Chicago** — M.S. in Statistics 2019 – 2021
Thesis: Novel Data Splitting Techniques for Post-Selection Inference in Poisson Regression
Advisor: Rina Barber
- The University of Chicago** — B.A. in Mathematics and Economics 2010 – 2013

PUBLICATIONS AND PREPRINTS

- James Leiner**, Aurélien Bibaut, Nathan Kallus, Aaditya Ramdas, and Koulik Khamaru. Adaptive inference for functionals of M -Estimands. Accepted at *40th Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- James Leiner**, Robin Dunn, and Aaditya Ramdas. Adaptive off-policy inference for M -estimators under misspecification. *arXiv:2509.14218*, 2025.
- James Leiner**, Brian Manzo, Aaditya Ramdas, and Wesley Tansey. Scalable causal structure learning via amortized conditional independence testing. *4th Conference on Causal Learning and Reasoning (CLearR)*, 2025. **(Oral)**
- James Leiner**, Boyan Duan, Larry Wasserman, and Aaditya Ramdas. Rejoinder to the Discussion of “Data fission: splitting a single data point”. *Journal of the American Statistical Association (JASA)*, 2025.
- James Leiner** and Aaditya Ramdas. Graph fission and cross-validation. *27th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
- James Leiner**, Boyan Duan, Larry Wasserman, and Aaditya Ramdas. Data fission: splitting a single data point. *Journal of the American Statistical Association (JASA)*, 2023. **(Discussion Paper)**

TALKS AND INVITED SESSIONS

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| IMS International Conference on Statistics and Data Science, <i>Adaptive inference for M-estimators</i> | December 2025 |
| Conference on Causal Learning and Reasoning, <i>Scalable causal structure learning</i> | May 2025 |
| Meta Central Applied Science Research Seminar, <i>Data fission and Gaussian Process regression</i> | January 2025 |
| Joint Statistical Meeting, <i>Data fission: splitting a single data point (discussion paper talk)</i> | August 2024 |
| International Seminar on Selective Inference, <i>“Inference after Latent Variable Estimation” discussant</i> | June 2022 |
| International Seminar on Selective Inference, <i>Data fission: splitting a single data point</i> | May 2022 |

RESEARCH INTERNSHIPS

- Meta**, Research Scientist Intern Summer 2026
- Developed synthetic control methods for estimating average treatment effect in advertising geo-experiments, including cross-validation for estimator/ hyperparameter selection and Gaussian process estimators
 - Redesigned donor selection using factor models and SVD to minimize estimator variance, reducing experimental-design runtime by 20× and ATT standard errors to roughly one-third of the previous approach
- Novartis Pharmaceuticals Corporation**, Biostatistics Intern Summer 2024
- Developed causal inference and off-policy evaluation methods for adaptively collected clinical trial data
 - *Publication*: Adaptive off-policy inference for M -estimators under misspecification (preprint)

- Developed causal discovery methods to identify genetic biomarkers predictive of cancer metastases
- *Publication:* Scalable causal structure learning via amortized conditional independence testing (**CLearR**)

INDUSTRY EXPERIENCE

Novantas (now Curinos), Engagement Manager

2014–2016, 2017–2019

Promoted Associate → Lead Associate → Senior Associate → Engagement Manager

(financial services management consultancy)

- Led quantitative consulting engagements (3–7 person teams) for retail and commercial banks across deposit pricing, treasury management fee pricing, stress testing, funds transfer pricing, asset/liability management, and credit risk
- Managed client and stakeholder relationships at regional and multinational banks in the U.S., Canada, and Australia
- Built SAS/Python tooling to automate development and validation of GLM, ARIMA, and Cox PH models
- Designed mortgage early warning systems for a Big Four Australian bank to identify loans at high risk of delinquency

Facebook (now Meta), Quantitative Analyst

2016

- Developed data infrastructure and analytical reporting systems supporting large-scale marketing campaigns

TEACHING EXPERIENCE

Graduate Teaching Assistant, Carnegie Mellon University

Deep Learning (46-937)

Fall 2026

Machine Learning Capstone Course (46-983)

Fall 2025

Natural Language Processing (46-924)

Fall 2024

Machine Learning I (46-926)

Fall 2024

Intermediate Statistics (36-705)

Fall 2023

ABCDE of Machine Learning (36-708)

Spring 2023

Statistical Machine Learning (35-708)

Spring 2022

Modern Regression (34-601)

Fall 2021

Teaching Assistant, University of Chicago

Applied Regression Analysis (BUSN 41100), Booth School of Business

Fall 2020

Pre-Calculus (MATH 10500)

Fall 2011

HONORS AND AWARDS

Runner-up Poster Award, 12th International Conference on Multiple Comparisons Procedures

Fall 2022

Departmental Scholarship, The University of Chicago

Fall 2019

General Honors, The University of Chicago

Fall 2013

Dean's List, The University of Chicago

2010–2013

National Merit Corporate Scholarship, Marsh and McLennan Companies

2010

PROFESSIONAL SERVICE

Reviewer: Journal of the American Statistical Association (3), Journal of the Royal Statistical Society Series B (4), Electronic Journal of Statistics, Biometrika, NeurIPS

COMPUTER SKILLS

Coding Python (NumPy, pandas, scikit-learn, PyTorch), R (tidyverse, ggplot2), SAS, SQL